Does the presence of linear kinematic features in the Arctic sea ice have an impact on the ocean surface?

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Key Points:

- Buoyancy fluxes in Linear Kinematic Features vary seasonally
- Buoyancy fluxes in Linear Kinematic Features can be 20-30% larger than in the ice-pack in summer.
- The presence of Linear Kinematic Features has a small impact in the transform water masses at the ocean surface.

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Abstract

The Arctic sea-ice, in particular the ice-pack, acts as an insulator between the atmosphere and the ocean below. Linear kinematic features (LKF), common in Arctic sea ice, facilitate exchanges of heat, salt, moisture and gases between the ocean and the atmosphere. Therefore, LKFs can influence heat exchanges, ice melt or growth, ocean dynamics, and the atmospheric boundary layer. The existing literature suggests that LKFs play a significant role in the heat budget of the surface of the ocean and likely impact water mass transformation within the Arctic. However, due to the lack of observations and high resolution models capable to resolve LKFs, an estimate of the impact of LKFs in the ocean has been elusive. Here, we explore the magnitude and impact in the ocean surface caused by the LKFs using SEDNA, a really high-resolution pan-Arctic hindcast (1/60°; NEMO + SI3 configuration). Despite LKFs accounting for less than 20% of the sea-ice cover area, they can contribute up to 10% more buoyancy than the ice-pack at the surface. Of the total surface water mass transformation in the ice covered Arctic, LKFs explain approximately 10%. The present estimate indicates that while LKFs impact the magnitude of the WMT, they make a small contribution to Arctic Ocean dynamics since, the buoyancy flux and water mass transformation in the LKFs are similar to those of the surrounding ice-covered areas, contrary to previous hypotheses.

1 Introduction

The Arctic sea-ice plays a crucial role in regulating Earth's climate by acting as a natural insulator between the atmosphere and the ocean below (Wettlaufer et al., 1997; Untersteiner, 1961). This insulating property acts as a thermal barrier, limiting heat loss from the ocean to the atmosphere and minimizing evaporation. It thereby maintains the stability of the polar environment (Ruffieux et al., 1995). A ubiquitous feature of Arctic sea ice is the formation and persistence of linear kinematic features (LKFs). LKFs are defined as the regions where the total deformation of sea ice reaches a local maxima (See methods for the definition of total deformation; Hutter et al. 2019). These features are characterized by sea ice shear and divergence that can result in large flux exchanges between the ocean and the atmosphere (Lüpkes et al., 2008). These deformations vary in width from a few meters to kilometers, with length-scale spanning distances of a few kilometers up to hundreds of kilometers (Tschudi et al., 1998). LKFs can be formed through the divergence or/and shear of the sea ice forced by winds and oceanic currents (Rampal et al., 2016; Hutchings et al., 2005; Richter-Menge et al., 2002). Winds are thought to be the main forcing of the sea ice deformation, where persistent winds stress result in the formation of LKFs (Linow & Dierking, 2017; Hutter et al., 2018). Although ocean currents have a smaller contribution to the sea ice deformation compared to wind forcing, ocean drag at the bottom of the ice acts to balance the wind-induced sea ice velocities (Weiss & Marsan, 2004). Additionally, observations show that ocean currents may play a bigger role in dictating the fracturing patterns of sea ice in regions with strong currents and bathymetric features (Willmes et al., 2023).

Once a LKFs are formed, air temperatures and localized upwelling can generate important heat fluxes at the ocean surface (McPhee et al., 2005; Marcq & Weiss, 2012; Maykut, 1986), instigating localized melting/freezing of sea-ice (von Albedyll et al., 2022). Additionally, changes in buoyancy forcing due to brine rejection into the mixed layer can induce convection (D. C. Smith & Morison, 1998), leading to the generation of fronts, mixed layer turbulence, and eddy generation (D. C. Smith et al., 2002). Furthermore, changes in the buoyancy could result in the local transformation the surface water masses (Pemberton et al., 2015; Walin, 1982). Therefore, literature suggests that LKFs may play a significant role in the Arctic Ocean by modulating stratification, generating instability, influencing ocean circulation, and transforming water mass (Ringeisen et al., 2023; Reiser et al., 2020; S. D. Smith et al., 1990). However, the magnitude and influence of LKFs compared to the surrounding ice cover in the Arctic basin are yet to be quantified. **@Camille: do you have in mind any other papers? I can't think of others to add more detail on why LKFs matter for ocean processes.**

In-situ observations have provided evidence that LKFs locally impact the ocean structure of the Arctic Ocean. However, the lack of observations limits an estimation of their integrated impact on the Arctic Ocean. Here, we present the first estimate of the impact of the LKFs in the Arctic Ocean, using a really high-resolution pan-Arctic hindcast, SEDNA (Talandier & Lique, 2023). Since the ice cover is dynamically and thermodynamically different in the marginal ice zone and the ice pack, we quantify the impact of LKFs for both regions using buoyancy fluxes and the water mass transformation framework (See Methods for details of the calculations and numerical model). Our analysis assesses the impact of LKFs in the buoyancy flux and water mass transformation of the Arctic Ocean, providing insight for assessing the evolution of the Arctic ocean underneath a sea ice cover.

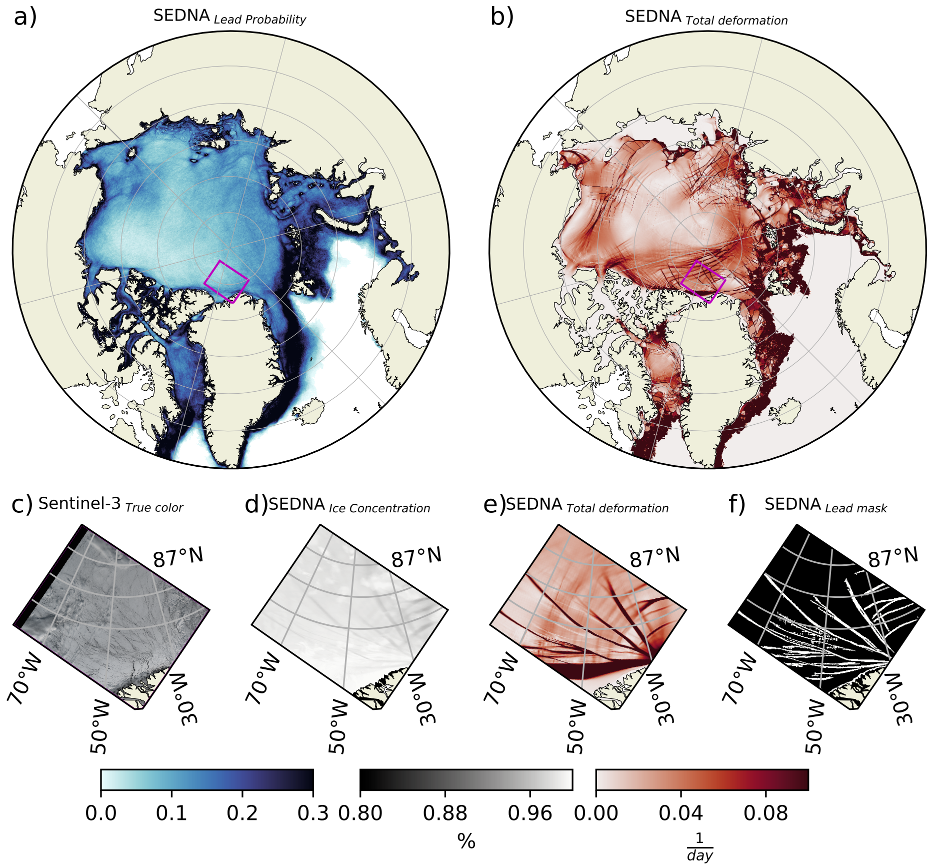

Figure 1. Linear kinematic features from SEDNA. a) Linear kinematic mean frequency during winter between 2011 and 2015. b) Snapshot of the total deformation for the 21st of April 2014. c) True color image from Sentinel 3 on the 12th of April 2022 for the magenta area in panel a. d) Mean ice concentration from SEDNA (800m resolution) on the 21st of April 2014. e) Total deformation used to identifies the linear kinematic features. f) Identified linear kinematic features on the 21st of April 2014.

Observation vs Models

Climate models and high resolution sea ice models are unable to reproduce LKFs due to the lack of resolution to capture their spatial scale (Wang et al., 2016). SEDNA is capable of overcoming this limitation and capture most of the LKF width distribution ($> 1km$; Wernecke & Kaleschke 2015), since the model resolution is $\sim 800m$ on average across the Arctic. The climatology of the LKF probability, i.e. the likelihood of a LKF occurring for any given day, is shown in figure 1a. The mean probability of LKFs averaged over the Arctic is $\sim 19\%$ in SEDNA, comparable to observations using RADARSAT

Geophysical Processor System and Thermal-Infrared Satellite Imagery (Wernecke & Kaleschke, 2015; Willmes et al., 2023). Although observations use different identification methods compared to the one implemented in SEDNA (Hutter et al., 2019), there is resemblance between the spatial patterns of SEDNA and observations (Supplementary Fig. 5). For example, we observe increased LKF frequency in the shelves, along bathymetric features, and in regions with large kinetic energy (Fig. 5). Several hotspots of LKF occurrence can be identified along the sea shelves and in the Chukchi Sea, highlighting the model's ability to reproduce LKFs dynamics. SEDNA reproduces the patterns observed in the Arctic shelves, the Greenland Sea, Barents Sea, Kara Sea and Laptev Sea, but the prominent signature in the Beaufort Sea in observations is not captured in SEDNA. **@Camille, you asked for something more quantitative, between the obs and the model. Do you have any specific idea?**

A snapshot of the total deformation is shown in figure 1b. A closer examination, focusing north of Greenland in panels c, d, e, and f of figure 1, reveals similarities between the spatial pattern of typically observed LKFs on the 12th of April 2022 from a true color satellite image (Fig. 1c) and the ice concentration of SEDNA for the 21st of April 2014. Note that the sea ice concentration in the identified LKFs shown in figure 1f using the LKF identification algorithm proposed by Hutter et al. (2019) remains higher than 80% concentration; this is a consequence of the visco-plastic nature of the sea ice model and the daily averaged output that smooths the presence of LKFs in the fields. Furthermore, in our simulation, but in fact, in all numerical models using a visco-plastic rheology, it has been observed that they do not reproduce the typical fracturing angles as in observations (Hutter et al., 2022). Despite this, the similarities between the observations and model LKF probability allow us for the first time to estimate quantitatively the impact LKFs have in the Arctic ocean.

In situ observations have shown that LKFs can have up to two orders of magnitude larger heat fluxes than the surrounding ice covered ocean (McPhee et al., 2005; Maykut & McPhee, 1995). Consequently, they could modify the fluxes at the ocean-atmosphere interface of the Arctic basin (Ólason et al., 2021). A change in these fluxes could impact the buoyancy flux and the circulation of the Arctic Ocean. Figure 2a and e show two snapshots of the buoyancy flux, with a negative buoyancy flux in winter resulting from sea ice refreezing and brine rejection. Meanwhile, a positive buoyancy flux in summer is associated with sea ice melting and freshening of the Arctic surface (Fig. 2b). Inside the

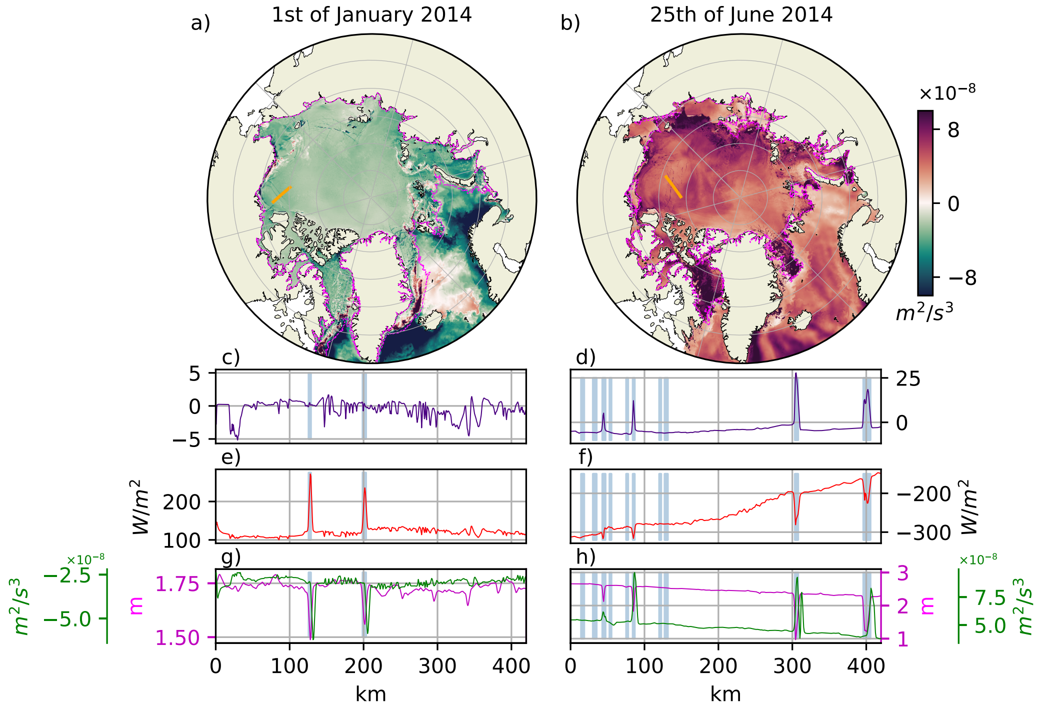

Figure 2. Maps of buoyancy flux and transects for the 1st of January and 25th of June 2014. a) Buoyancy flux in Winter the 1st of January 2014. Transect on the 1st of January 2014 of c) downward heat flux, e) freshwater heat-equivalent flux, g) ice thickness and mixed layer depth. b) Buoyancy flux in Summer the 25th of June 2014. Transect on the 25th of June 2014 of d) downward heat flux, f) freshwater heat-equivalent flux, h) ice thickness (magenta) and buoyancy flux (green). Note that the transect location and orientation changes between the winter and summer dates. Magenta contour in panels a and e correspond to the 0% concentration contour of sea ice. Vertical blue bars shows the mask of LKFs in the transect.

ice covered region (magenta contour) in figure 2, the fluxes are spatially homogeneous, except for the marginal ice zone and the LKFs, where the buoyancy fluxes are larger than those of the surrounding ice cover ocean, in agreement with previous studies. The flux across the Arctic for the LKF is $-2.46 \times 10^{-8} m^2/s^3$ and $7.20 \times 10^{-8} m^2/s^3$ for each date, respectively, compared to the $-2.42 \times 10^{-8} m^2/s^3$ and $5.89 \times 10^{-8} m^2/s^3$ of the surrounding ocean. In other words, in winter, the fluxes across the LKF and the surrounding ocean are comparable. However, in summer, the fluxes in the LKFs can be up to 20% larger than those in the ice covered ocean.

Two transects across the Beaufort Gyre for the 1st of January 2014 and 30th of June 2014 are highlighted by the orange lines in the maps of Figure 2. Panels b and f display the heat flux, panels c and h show the freshwater heat-equivalent flux (equivalency $1W/m^2 = 1{\times}10^-6kg/m^2s^1$), panels d and h present the ice thickness and buoyancy flux, and vertical blue bars indicate the identified LKFs along the transects. Examining the 1st of January 2014, we observe a small heat flux in the LKFs (Fig.2 b) ,which does not impact the heat flux at the ocean-atmosphere interface since the ocean surface is at freezing point. In other words, the extra heat exchanged with the atmosphere is used by the sea ice model to grow sea ice instead of the ocean model, resulting in a heat flux close to zero at the ocean-atmosphere interface. In contrast, the salt flux (Fig.2e) shows two positive peaks, each with twice the magnitude of the surrounding ice cover, collocated within the identified LKFs. This is due to an enhanced sea ice growth and brine rejection within the LKFs. The salt flux and heat flux result in a negative buoyancy flux for this snapshot where the largest buoyancy loss occurs within the LKFs (Fig.2 g). Additionally, within the LKFs there is a decrease in the sea ice thickness of $25cm$. On the 25th of June (Fig.2d), we observe an increase in the heat fluxes within some of the identified LKFs, reaching up to $\sim 25W/m^2$. Meanwhile, the freshwater flux shows a local decrease within the LKFs, indicating a freshening of the surface due to sea ice melt. This change is captured by the positive buoyancy flux, decreasing the density at the surface (Fig.2f and h). The enhanced sea ice melt in the LKFs results in a sea ice thickness decrease of more than $1m$ within the LKFs. It is important to note that some peaks in the buoyancy flux align with the edge of the detected LKFs, likely due to the enhanced refreezing or melting that occur at the edge of the LKFs. Of all the identified LKFs, some exhibit significant ocean-atmosphere fluxes, while others have little to no imprint in the fluxes. LKFs without large ocean-atmosphere fluxes may be explained by the temporality of the sea ice break up (divergence), where later snapshots show an increase of fluxes at the location of identified LKFs. Alternatively, LKF generated through shear and convergence have a limited flux exchange with the atmosphere, but may have enhanced fluxes in the ocean interior due to isopycnal upwelling forced by the ice-ocean stress.

Similar to observations, we identify fluxes that are larger in the LKFs than in the surrounding ice pack in both summer and winter. However, the freshwater heat-equivalent flux, attributable to melting and freezing, is significantly larger than the heat fluxes associated with the LKFs. In the next section, we explore the magnitude of the buoyancy fluxes at the ocean-atmosphere interface averaged over the Arctic ocean during 2014. This analysis is conducted separately for the marginal ice zone (MIZ; 0-80% sea ice concentration), ice pack (80-100% sea ice concentration) and the LKFs within each of these regions.

LKF contribution to the ocean buoyancy

In the Arctic ice covered ocean, the averaged percentage area covered by four distinct masks representative of the sea ice covered Arctic Ocean during 2014 is as follows: 18.7% for the MIZ, 81.3% for the ice pack, 14.41% for the LKFs in the ice pack, and 10.04% for the LKFs in the MIZ. The area covered by each mask exhibit a seasonal cycle (Fig. 3a). During summer (June, July, and August), the percentage of the ice pack cover area decreases from approximately 90% to 50%, while the MIZ increases from 10% to approximately 50%. On one hand, the percentage of the area covered by LKFs in the ice pack peaks in winter (December, January, and February) at almost 20% coverage over the ice pack area. On the other hand, the LKFs in the MIZ peak in summer, covering up to 25% of the total ice cover Arctic and 50% of the MIZ surface area. The maxima LKFs area coverage is $\sim$ 20% of the total sea ice covered Arctic, consistent with previous model assessments (Wang et al., 2016).

To quantify the contributions of LKFs to ocean buoyancy, we integrate the buoyancy flux and area weight them for each of the sea ice cover masks. The buoyancy flux shows a significant seasonal cycle, with positive values corresponding to the sea ice melting season (decrease in buoyancy) and negative values coinciding with the sea ice freezing season (increase of buoyancy). During winter in the MIZ, the mean buoyancy flux is -1.20$\times 10^{-14} m^2/s^3$, with a haline component of 1.67$\times 10^{-14} m^2/s^3$, and a thermal component of -2.87$\times 10^{-14} m^2/s^3$. In summer, the mean buoyancy flux is 9.68$\times 10^{-14} m^2/s^3$, with a haline component of 9.42$\times 10^{-14} m^2/s^3$ and a thermal component of 0.26$\times 10^{-14} m^2/s^3$. In the MIZ, the buoyancy flux suggests a preferential refreezing of sea ice during winter and melting in summer (Fig.3b), since thermal component is the dominant term and negative during winter and the haline component is the dominant term and positive during summer (Fig.3c). In the ice pack, the buoyancy flux also changes sign from winter to summer (Fig.3b), indicating a shift from ice formation in winter to ice melting in summer, primary driven by the haline buoyancy component. In the ice pack, the mean winter buoyancy flux is -3.30$\times 10^{-14} m^2/s^3$, with a haline component of -3.24$\times 10^{-14} m^2/s^3$

and a thermal component of -0.06$\times 10^{-14} m^2/s^3$. In summer, the mean buoyancy flux is 5.92$\times 10^{-14} m^2/s^3$, with a haline component of 5.92$\times 10^{-14} m^2/s^3$ and a negligible thermal component. In both regions, the buoyancy flux in the LKFs follow the overall seasonal cycle of the buoyancy fluxes of the surrounding sea ice covered ocean. However the winter LKFs in the MIZ and the summer LKFs in the ice pack exhibit larger buoyancy fluxes compared to those in the MIZ and ice pack, respectively. The mean winter buoyancy flux of the LKFs in the MIZ is -1.98 $\times 10^{-14} m^2/s^3$, representing a 40% larger buoyancy flux in winter within the LKFs compared to those in the MIZ, suggesting enhanced freezing within the LKFs. The mean summer buoyancy flux of LKFs in the ice pack is 7.26$\times 10^{-14} m^2/s^3$, approximately 20% more buoyancy flux in summer compared to the ice pack, indicating enhance melting with in the LKFs.

@Camille: Should we add the following paragraph about MLD changes (See Supplementary Fig 6) As proposed by McPhee et al. 2005, LKFs generated by shear can significantly influence in the ocean MLD, by upwelling the isopycnals. Thus, to comprehensively understand the impact of LKFs on the ocean, we investigate the mixed layer depth, referenced to the first ocean model level, for each of the ice covered regions (Supplementary Fig.6). The MLD in the MIZ reaches a maximum depth in winter of 130m and a minimum MLD of 1.62m in summer. The MLD of the MIZ LKF is comparable to that of the MIZ, reaching 127m in winter and 1.67m in summer. Meanwhile, the MLD in the ice pack and the ice pack LKFs are deeper in winter (40.5m and 47.7m, respectively) and shallower in summer (1.9m and 1.7m, respectively). The difference in MLD between the ice pack and ice pack LKF is significant and, on average, the MLD is $\sim$ 20% shallower during winter in the ice pack LKFs. This results provides evidence that ice pack LKFs can impact MLD thickness through upwelling.

To further quantify the role and impact of LKFs in the Arctic circulation, we use the water mass transformation framework (WMT; Walin, 1982). This approach provides information on the formation of lighter or denser water due to changes in the surface fluxes. In figure 4a, the averaged WMT during 2014 is shown for the ice covered Arctic, the ice pack, the MIZ, and the LKFs for each of these areas. It is important to note that the ice covered Arctic is the sum of the ice pack and MIZ. The yearly mean WMT in the ice pack is characterized by a broad region of strong positive transformation (losing buoyancy) in the density range of $23 < \gamma_n < 27.5$, reaching a maximum of $\sim$ 2.1 Sv. Additionally, there is a small negative transformation (gaining buoyancy) below $\gamma_n < 23$.

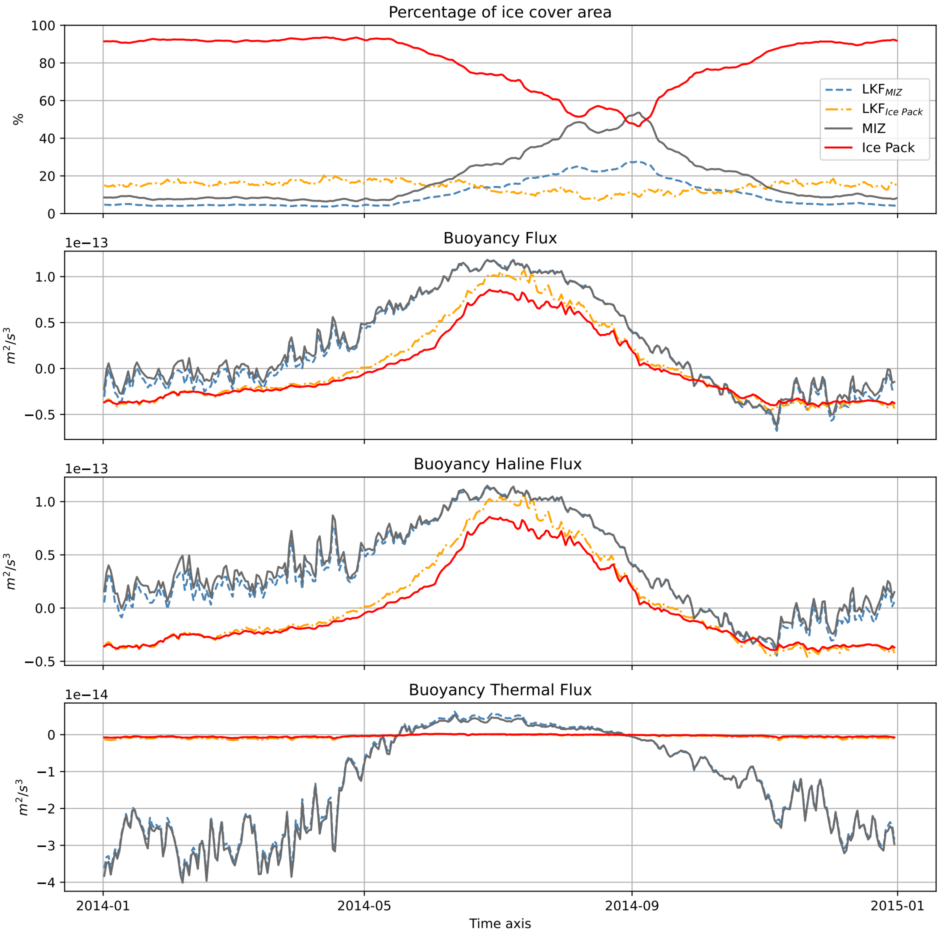

Figure 3. Time-series during 2014 of a) the area cover, b) buoyancy fluxes, c) haline buoyancy flux component, and d) thermal buoyancy flux component for linear kinematic features in the ice pack, the linear kinematic features in the marginal ice zone, the marginal ice zone (ice concentration 0-80%), and the ice pack (ice concentration >80%).

The yearly mean WMT in the ice pack LKFs also features a positive peak within the $24.5 < \gamma_n < 27.5$ reaching up to 0.39 Sv. On average the WMT in the ice pack LKFs accounts for $\sim 10\%$ of the WMT of the ice pack. The MIZ mean WMT is mostly negative within the density range of $20 < \gamma_n < 27.5$ (gaining buoyancy), reaching a minimum of -1.2 Sv around $\gamma_n = 27$. The MIZ LKFs WMT is also negative within the same range of densities, with a minimum of -0.6 Sv. On average, the WMT in the MIZ LKFs corresponds to $\sim 50\%$ of the WMT of the MIZ.

A Hovmoller diagrams of the WMT for the ice covered Arctic, the ice pack, the MIZ, and LKFs in each of these regions are depicted in figure 4. The ice covered WMT exhibit an important seasonal cycle with a mean positive WMT during winter of 1.37 Sv and mean negative WMT of -2.38 Sv during summer (4 b). The ice covered LKF WMT also shows a seasonality, where the WMT is 0.27 Sv in winter and -0.85 Sv in summer (4 c). LKFs in the ice covered Arctic contribute approximately 23% of the WMT throughout the year. This contribution of LKFs can be decomposed into their contribution in the ice pack and the MIZ allowing us to better understand the evolution of the WMT. In winter, a positive transformation occurs (loss of buoyancy), due to brine rejection due to sea ice growth. In summer, a negative transformation or a gain of buoyancy occurs due to the melting of sea ice. LKFs explain 10% of the WMT in the ice pack, however, there is not a emergence of a pattern in the WMT ratios suggesting that LKFs transform different water masses than the water masses being transformed underneath the ice pack (4 g). Second, the MIZ and the LKFs in this region present also a seasonal cycle, but the WMT is generally negative suggesting a gain of buoyancy due to melt of sea ice. LKFs in the MIZ explain approximately 50% of the MIZ WMT, and similarly to the LKFs in the ice pack, no pattern emerges from the WMT rations that suggest a specific role of LKFs in the WMT compared to the WMT of the MIZ.

@Camille: Should we add the following paragraph about water mass formation (See panel K in Fig 4)

In figure 4k, the formation rate integrated in $0.5kg/m^3$ density bins for the different ice covered masks, and the LKF within each area. The formation rate is the derivative of the mean WMT with respect to the density and it corresponds to the convergence of volume in density space. A negative formation in the upper ocean over a density range indicates that water must be upwelled from the ocean interior or destructed into a different water mass. We note that the sign of the water mass formation for each region is consistent with the water mass formation of the surrounding ice covered regions. There is a notable destruction of water in the PDW ($27.41kg/m^3 < \gamma_n < 27.81kg/m^3$) dominated by the water mass formation in the MIZ and the MIZ LKFs. This is likely a consequence of ice freezing and melting making a dominant contribution to the destruction in the regions where the PDW outcrops within the MIZ. In contrast, AW ($27.81kg/m^3 < \gamma_n$) and part of the PSW$_H$ ($26kg/m^3 < \gamma_n < 27.41kg/m^3$) formation occurs principally in the ice pack, with little contribution from the LKFs. Finally, the formation/destruction of other water masses (e.g., PSW_{ML} and lower branch of PSW$_H$) are dominated by the non-LKFs regions.

2 Conclusions

Linear kinematic features are ubiquitous in the sea ice covered oceans that allow exchanges between the ocean and the atmosphere. Generally, the importance of these fluxes at the ocean-atmosphere interface are considered to be of significant importance for the Arctic dynamics (Maykut & McPhee, 1995). The present study assesses the impact of LKFs in the ocean surface by diagnosing the buoyancy fluxes using high-resolution hindcast (SEDNA). The LKF identification algorithm implemented in SEDNA allows us to reproduce the LKFs spatial distribution over most regions of the Arctic, with exception of the Beaufort Sea. Despite these regional differences, SEDNA can estimate the mean LKF probability, showcasing its capacity to capture the dynamics of these features. This high resolution simulation allows us to better understand the role of LKFs in shaping the Arctic ocean.

Using SEDNA, a high-resolution pan-Arctic hindcast, we found that LKFs, although covering less than 20% of the sea-ice, can contribute up to 10% more buoyancy fluxes at the ocean surface than the ice-pack. They also account for approximately 20% of the total surface water mass transformation in the ice covered Arctic. Contrary to previous hypotheses, our findings suggest that buoyancy forcing through LKFs has a smaller contribution to Arctic Ocean dynamics than initially thought. While their presence can induce localized changes in the density of surface waters due to brine rejection and ice melting, their spatial extent and duration are not sufficient to significantly alter the surface water masses. Furthermore, both the transformation and formation of water masses in the LKFs have the same sign, and they are comparable to those in the surrounding ice covered areas. Thus, there is no evidence that LKFs act differently or that they have a particular imprint in the Arctic ocean water masses.

We observe a lifting of the MLD in the ice pack LKFs, likely linked to the ice-ocean stress in LKFs generated by shear, as previously discussed by McPhee et al. 2005. Although, the MLD thickness is $\sim 20\%$ shallower during winter in the ice pack LKFs, there is no evidence that LKFs experience a distinct response to the surface buoyancy fluxes. Thus, the remaining processes to be investigated are the impact of entrainment in the MLD within the LKFs, as well as the source of potential energy and generation of instabilities due to isopycnal tilting in the vicinity of LKFs.

@Camille, Is there any other point that we should discuss in the conclusions?

@Camille, No need to check the methods yet.

3 Methods

Model description

Here we employ the NEMO numerical platform (release 4.0.5) to establish a 1/60° pan-Arctic ocean-sea ice configuration named SEDNA. The ocean core (NEMO-OCE) solves the primitive equations (PE) using finite differences on an Arakawa C-grid mesh, while the ocean is coupled to the SI3 sea-ice component (NEMO-SI3). The pan-Arctic domain covers the Bering Strait on the Pacific side and extends southward to 56°N in the Subpolar North Atlantic gyre and to the Baltic Sea entrance. The ocean model incorporates a linear free surface, utilizes a 3rd order flux form scheme for momentum advection, and employs a 4th order Flux Corrected Transport (FCT) for tracer advection. Additionally, the sea-ice representation includes a 5-category model with the Elasto-Visco-Plastic (EVP) rheology activated and a land-fast capability to better simulate sea-ice behavior in specific areas.

The NCAR bulk formula from Large & Yeager (2009) to calculate air-sea fluxes and uses hourly ERA5 atmospheric data for near-surface variables. The three lateral boundaries are constrained with daily GLORYS12V1 Re-analysis data, and monthly freshwater fluxes from river discharges are combined with Greenland land ice melt data to supply SEDNA. The simulation extends over seven years (2009 to 2015) and starts from rest, utilizing initial conditions based on the World Ocean Atlas 2009 temperature/salinity and mean January 2009 ice state from PIOMAS re-analysis. Preliminary assessment against available observations and re-analyses data shows that the simulation reasonably captures key ocean and sea-ice characteristics.

Identification of LKFs

The LKFs from SEDNA were identified by using the algorithm proposed by Hutter et al. (2019). This method uses the total deformation rate defined as:

$$T_d = \left[\left(\underbrace{\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}}_{Divergence} \right)^2 + \underbrace{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2}_{Shear} \right]^{1/2}, \tag{1}$$

where u and v are the sea ice velocities. The deformation rate varies spatially depending on the background deformation and the sea ice properties (), where the largest deformation rates occur along the boundaries of ice flows where the LKFs are found. Therefore, to identify the LKFs it is required to identify the local maxima using edge detection (DoG filter) in the deformation rate field. After the edge detection, using binary map we obtain a mask with the original width of identified LKF. This mask is then used to compute the statistics presented in this study. After this step, Hutter et al. (2019) reduce the LKFs width to a single pixel to then track the LKFs over time. Since our study focus on the ocean-atmosphere interactions, we only use the identification algorithm to produce a mask of LKFs during 2014 for the Arctic, where we retain the original width computed from the deformation rate. This mask is then used to quantify and contrast the impact of LKFs compared to the ice-pack.

Buoyancy flux and water mass transformation

- Describe MLD definition
- Describe buoyancy flux and decomposition between thermal and haline components.

4 Open Research

Acknowledgments

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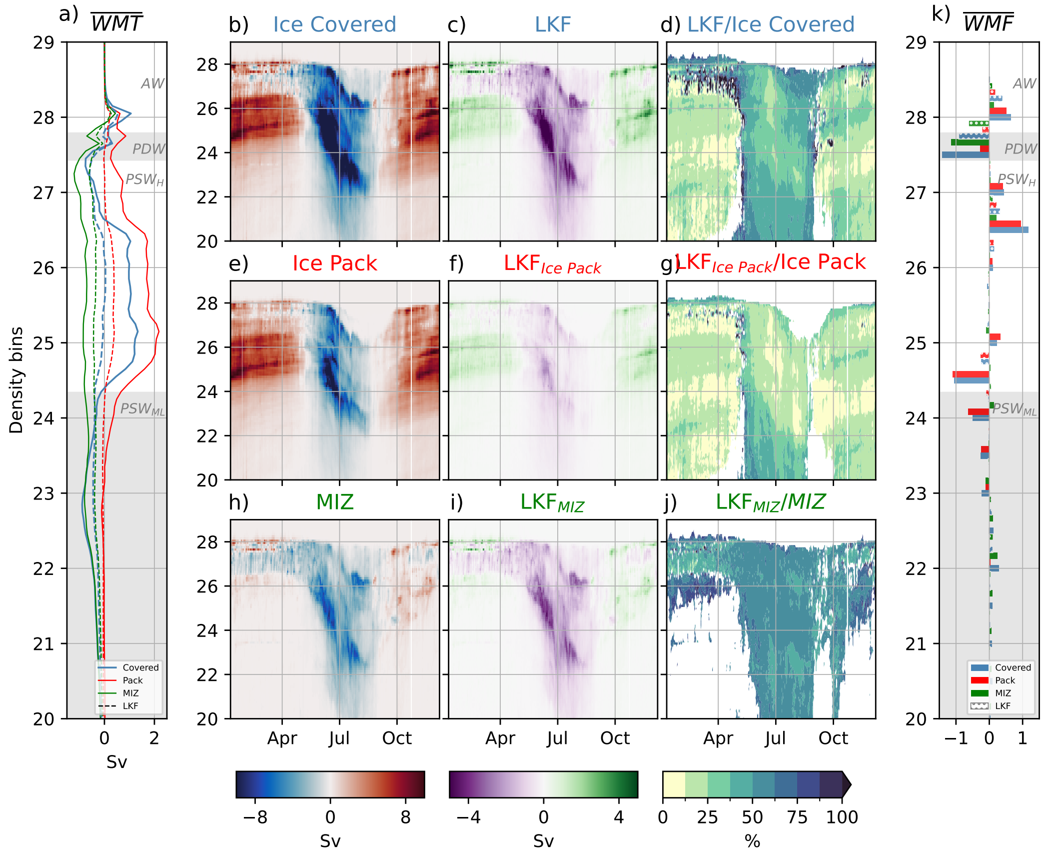

Figure 4. Mean and daily Water mass transformation over 2014. a) Mean water mass transformation over 2014 for the ice covered, ice pack, MIZ and LKF in each region. Hovmoller diagram of the water mass transformation (Sv) for the b) ice covered, c) ice covered linear kinematic features and d) the ratio of water mass transformation between the LKFs and the ice covered region. Panels e), f), and g), and h), i), and j) follow the same structure as panels b), c) and d), but for the ice pack and MIZ regions, respectively. k) Mean water mass formation in density bins of $0.5 kg/m^3$ during 2014 for the same areas as in panel a). Note the magnitude change in the colorbar of panels a), b), and c). Shaded gray and white areas in panels a) and k) represent the water masses of the Arctic ocean: the mixed layer polar surface water (PSW_{ML}; $\gamma_n < 24.36 kg/m^3$), the halocline polar surface water (PSW_H; $24.36 kg/m^3 < \gamma_n < 27.41 kg/m^3$), the polar deep water (PDW; $27.41 kg/m^3 < \gamma_n < 27.81 kg/m^3$), and the Atlantic water (AW; $27.81 kg/m^3 < \gamma_n$).

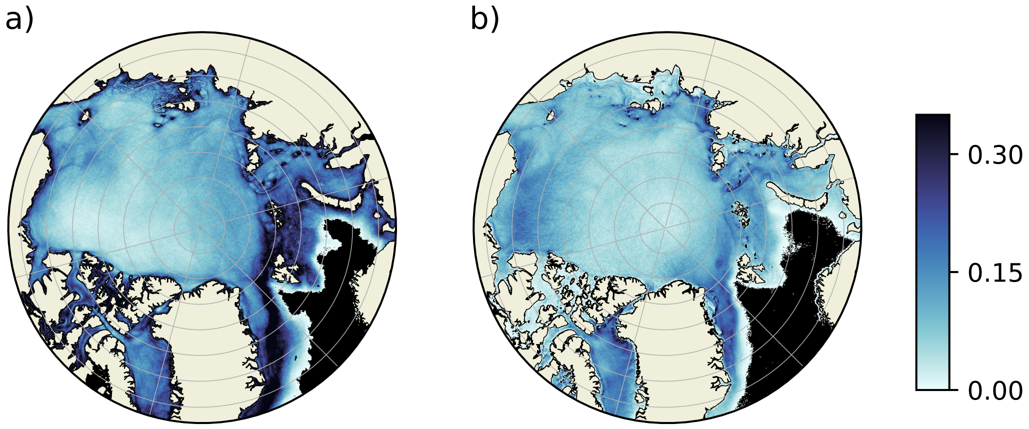

Figure 5. Linear kinematic mean frequency from a) SEDNA and b) Thermal-Infrared Satellite Imagery (Willmes et al., 2023) between 2011 and 2015 **TODO: Add contours of KE and topography**

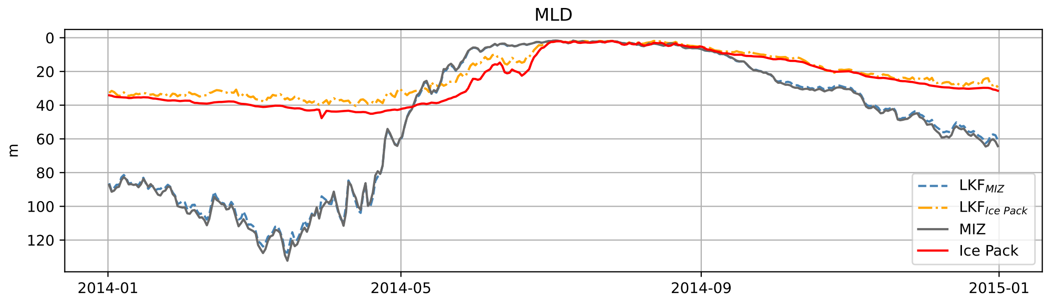

Figure 6. Time-series during 2014 of the mean mixed layer depth for linear kinematic features in the ice pack, the linear kinematic features in the marginal ice zone, the marginal ice zone (ice concentration 0-80%), and the ice pack (ice concentration ¿80%).